

Smart ATM Routing System (SARS)

Overview: The Smart ATM Routing System is a real-time, location-based integration platform designed to enhance the banking customer experience. It shifts the traditional ATM locator paradigm from simply finding the nearest geographical machine to identifying the nearest *available* and *operational* machine that provides the specific service the customer needs (e.g., cash withdrawal, deposit, currency exchange).

When an ATM goes out of service or runs out of cash, the system instantly detects the status change. If a customer attempts a transaction at a downed machine, the system dynamically calculates the best alternative ATMs based on real-time traffic, distance, and machine status. The customer is then presented with alternative locations directly on the ATM screen and is given the option to receive turn-by-turn navigation details via SMS. The system also includes a centralized web dashboard for bank administrators to monitor the health and cash levels of the entire ATM network in real time.

Functional Requirements (FRs)

These define the specific behaviors and functions the system must perform.

1. Real-Time ATM Monitoring & Status Updates

- The system must receive and process real-time status payloads (heartbeats) from ATMs across the network.
- The system must update the operational status (Online, Offline, Maintenance) and cash availability (Available, Low, Empty) for each machine.
- The system must track the specific services available at each ATM node (Withdrawal, Deposit, Currency Exchange, Check Deposit).

2. Smart Routing & Alternative Location Engine

- The system must capture the exact geographic coordinates (Latitude/Longitude) of the out-of-service ATM.
- The system must filter the database to find operational ATMs within a predefined radius that support the specific service the customer requested.
- The system must integrate with a mapping service (e.g., Google Maps Distance Matrix API) to calculate the exact distance and Estimated Time of Arrival (ETA) for the top 3 alternative ATMs.

3. Customer Interface (ATM Screen Integration)

- The system must expose an API endpoint to feed data to the ATM's frontend interface when a failure occurs.
- The system must display the 3 nearest operational alternatives, showing the branch/location name, distance, and ETA.
- The system must provide an input field on the ATM screen for the customer to enter their mobile phone number.

4. SMS Notification System

- The system must securely process the customer's phone number without storing it permanently.
- The system must dispatch an SMS containing the alternative ATM's address, requested service confirmation, and a direct Google Maps routing link.
- The SMS dispatch process must run asynchronously in the background to avoid blocking the ATM's interface.

5. Admin & Monitoring Dashboard

- The system must provide a secure web application for bank administrators.
- The dashboard must feature a live interactive map plotting all ATMs, utilizing color-coded markers (e.g., Green = Operational, Yellow = Low Cash/Warning, Red = Out of Service).
- The system must push state changes to the admin dashboard instantly without requiring a page refresh.
- The system must allow admins to filter ATMs by status, location, or supported services.

Non-Functional Requirements (NFRs)

These define system attributes such as performance, security, and scalability—critical for financial sector compliance.

1. Performance & Latency

- **Real-time Processing:** State changes from the ATM must be reflected on the Admin Dashboard within 2 seconds (utilizing WebSockets).
- **Routing Response Time:** The backend must calculate and return the alternative ATMs to the screen in under 3 seconds to prevent customer abandonment.

- **Asynchronous Execution:** API calls to external services (Google Maps, Twilio) must be handled via background task queues (e.g., Celery) to ensure main thread availability.

2. Scalability

- The backend architecture must be capable of handling concurrent heartbeat requests from thousands of ATMs simultaneously.
- The system must be stateless where possible to allow horizontal scaling of the API servers.

3. Reliability & Availability

- The system must target an uptime of 99.99% (High Availability).
- In the event of a mapping API failure (e.g., Google Maps downtime), the system must gracefully degrade and fall back to calculating direct point-to-point (Haversine) distance using local database coordinates.

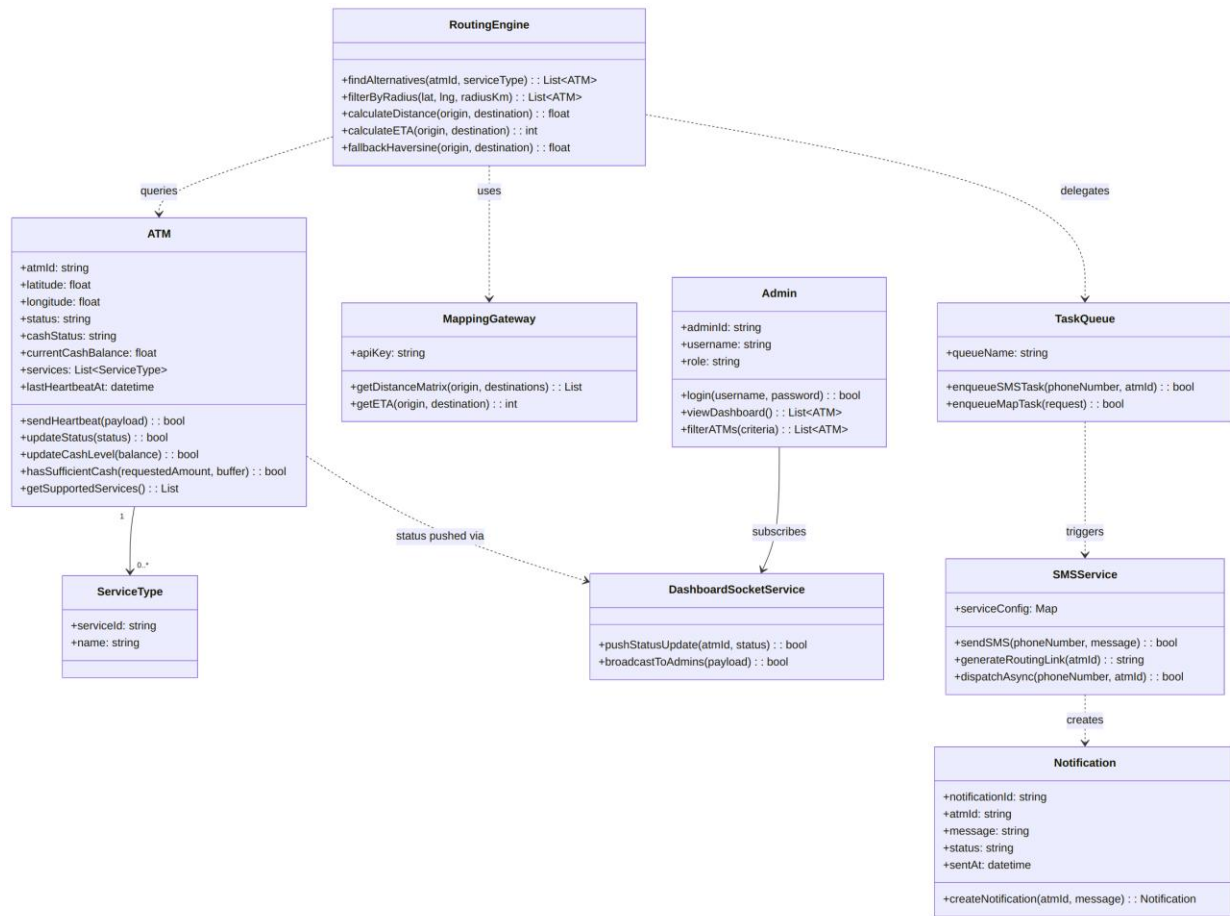
4. Security & Privacy

- **Data Protection:** Customer phone numbers entered into the ATM must be encrypted in transit (TLS 1.3) and must not be logged or persisted in the database after the SMS is dispatched (Data Minimization).
- **Authentication:** The Admin Dashboard must require strong authentication (OAuth2 or JWT) and enforce Role-Based Access Control (RBAC).
- **API Security:** All endpoints communicating with the ATMs must use API keys and IP whitelisting to prevent unauthorized access.

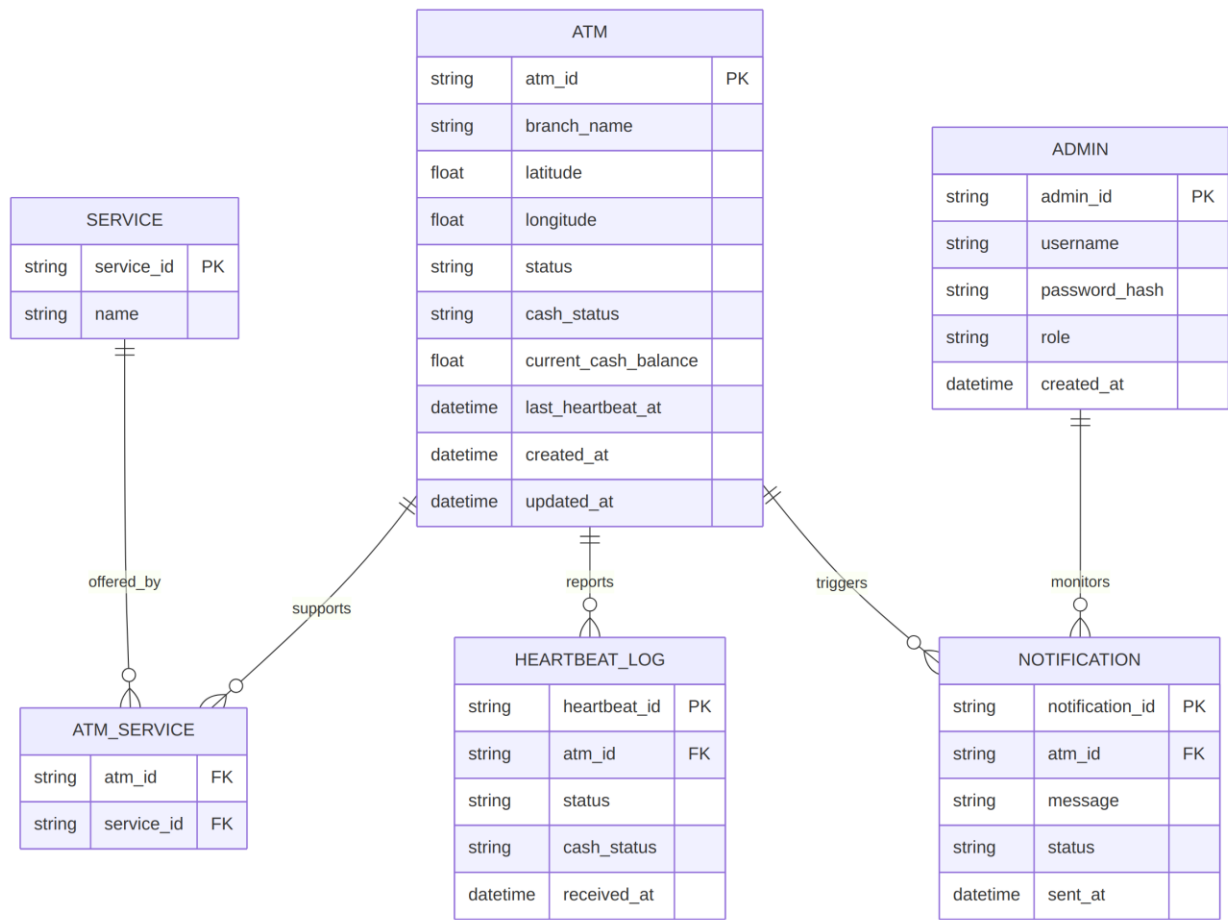
5. Maintainability

- The system must be fully containerized (Docker) to ensure parity across development, staging, and production environments.
- The codebase must follow a modular architecture (e.g., separating routing logic, notification logic, and socket management) to facilitate automated testing and CI/CD pipelines.

class diagram

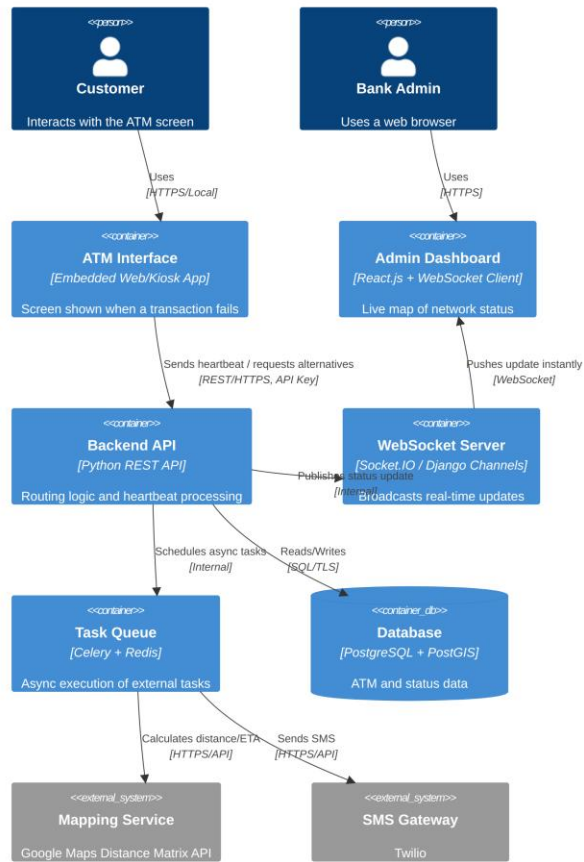


ER Diagram

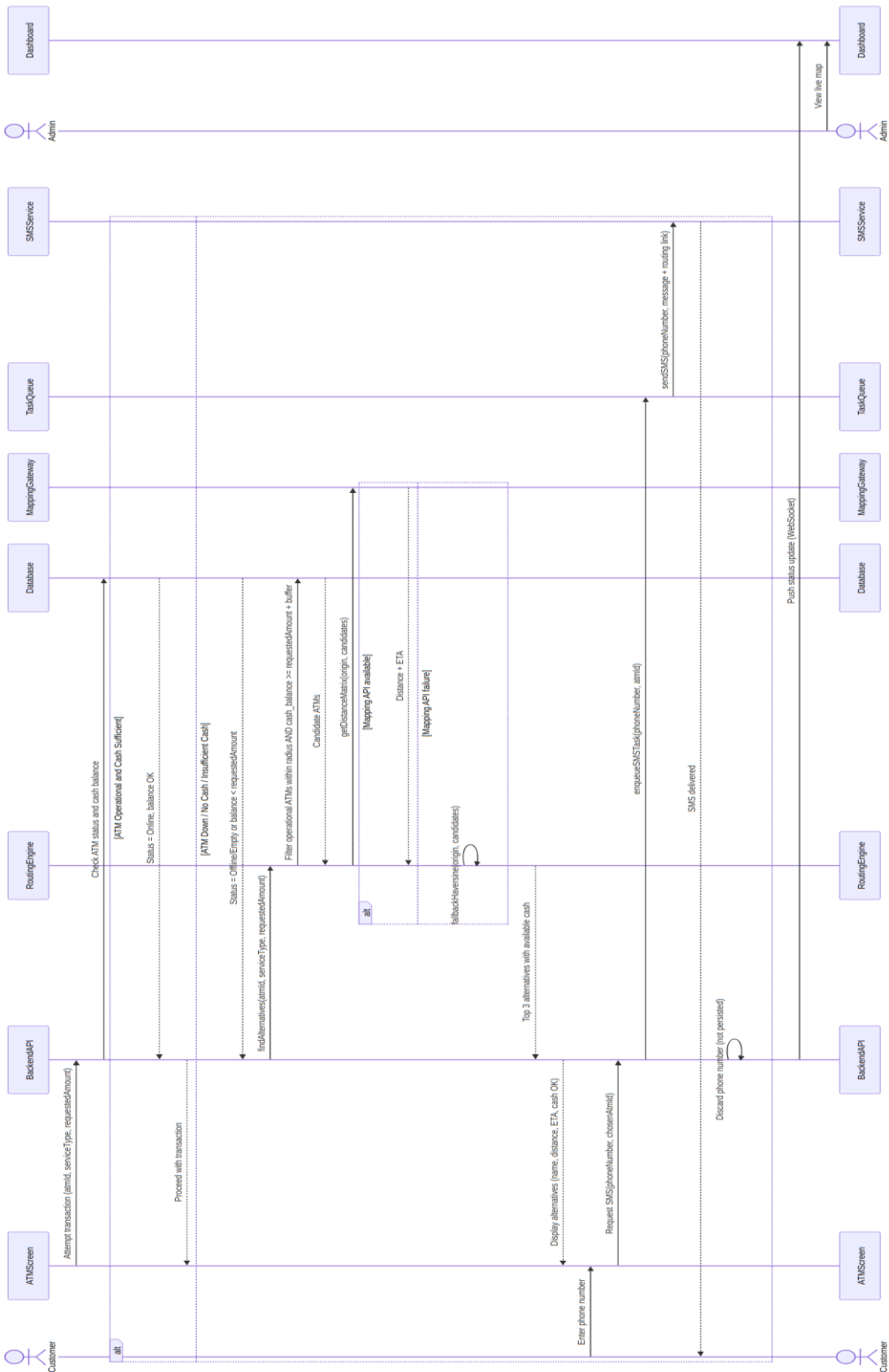


C4 Diagram

Container Diagram - Smart ATM Routing System



Sequence diagram



Full API Contract

Base URL: <https://api.sars.bank.com/api/v1>

1. Endpoint Summary

#	Method	Endpoint	Consumer	Auth	Purpose
1	POST	/atms/{atmId}/transactions/attempt	ATM screen	API Key	Attempt a transaction; returns proceed or alternatives
2	POST	/atms/{atmId}/heartbeat	ATM device	API Key	Report real-time status/cash/services
3	POST	/routing/alternatives	ATM screen	API Key	Get top 3 alternative operational ATMs
4	POST	/notifications/sms	ATM screen	API Key	Queue SMS with routing details
5	GET	/notifications/sms/{taskId}/status	ATM screen	API Key	Poll SMS delivery status
6	GET	/notifications	Admin	JWT	List/filter sent notifications (audit view)
7	POST	/auth/login	Admin	None	Authenticate admin, issue tokens
8	POST	/auth/refresh	Admin	Refresh token	Refresh access token
9	GET	/atms	Admin	JWT	List/filter all ATMs
10	GET	/atms/{atmId}	Admin	JWT	Get single ATM detail + heartbeat history
11	POST	/atms	Admin (SUPER_ADMIN)	JWT	Register new ATM
12	PATCH	/atms/{atmId}	Admin (SUPER_ADMIN)	JWT	Update ATM metadata
13	GET	/services	Admin / ATM	JWT / API Key	List supported service types

#	Method	Endpoint	Consumer	Auth	Purpose
14	WS	/ws/dashboard	Admin	JWT (query param)	Real-time push of ATM status changes

2. Common Headers

Header	Used On	Value
Authorization	Admin endpoints	Bearer <jwt_token>
X-API-Key	ATM endpoints	<atm_device_api_key>
Content-Type	All	application/json

3. Standard HTTP Status Codes

Code	Meaning
200	Success
201	Created
202	Accepted (queued for async processing)
400	Bad Request / validation error
401	Unauthorized
403	Forbidden (RBAC violation)
404	Not Found
422	Unprocessable Entity
429	Rate limited
500	Internal Server Error

Code	Meaning
503	Downstream service unavailable

4. Standard Error Object

Field	Type	Description
<code>error.code</code>	string	Machine-readable error code
<code>error.message</code>	string	Human-readable description
<code>error.details</code>	object/null	Extra context (e.g., validation errors)

5. POST /atms/{atmId}/transactions/attempt

Entry point called by the ATM screen whenever a customer starts a transaction. The backend checks the current ATM status and either tells the screen to proceed normally or triggers the alternative-routing flow in the same response. (Matches the first two steps of the Sequence Diagram: "Attempt transaction" -> "Check ATM status".)

Auth: X-API-Key **NFR:** must resolve in under 3s (same budget as the routing calculation it may trigger).

Path Params

Name	Type	Required	Description
atmId	string	Yes	The ATM where the customer is standing

Request Body

Field	Type	Required	Description
requestedService	string	Yes	WITHDRAWAL, DEPOSIT, CURRENCY_EXCHANGE, CHECK_DEPOSIT
requestedAmount	float	Required when requestedService = WITHDRAWAL	Amount the customer wants to withdraw

Response 200 OK — outcome: PROCEED

Field	Type	Description
outcome	string	"PROCEED"
atmId	string	Echoed ATM id

Response 200 OK — outcome: REDIRECT

Field	Type	Description
outcome	string	"REDIRECT"

Field	Type	Description
reason	string	OFFLINE, MAINTENANCE, NO_CASH, or INSUFFICIENT_CASH
requestedAmount	float	Echoed back for the ATM screen
alternatives[]	array<object>	Same shape as POST /routing/alternatives response, pre-filtered by cash sufficiency

Errors

Code	Reason
401	Invalid or missing API key
404	Unknown ATM id
503	Routing engine unavailable while resolving alternatives

6. POST /atms/{atmId}/heartbeat

Receives real-time status payloads from ATMs. (FR 1)

Auth: X-API-Key **NFR:** must be reflected on the dashboard within 2s.

Path Params

Name	Type	Required	Description
atmId	string	Yes	Unique ATM identifier

Request Body

Field	Type	Required	Allowed Values	Description
status	string	Yes	ONLINE, OFFLINE, MAINTENANCE	Operational status
cashStatus	string	Yes	AVAILABLE, LOW, EMPTY	Cash availability (derived label)
cashBalance	float	Yes	≥ 0	Actual cash loaded, persisted as current_cash_balance
services	array<string>	Yes	WITHDRAWAL, DEPOSIT, CURRENCY_EXCHANGE, CHECK_DEPOSIT	Supported services
latitude	float	Yes	-90 to 90	Geo coordinate
longitude	float	Yes	-180 to 180	Geo coordinate
timestamp	ISO8601 string	Yes	—	Time the payload was generated

Response 200 OK

Field	Type	Description
atmId	string	Echoed ATM id

Field	Type	Description
received	boolean	Confirms processing
processedAt	ISO8601 string	Server-side timestamp

Errors

Code	Reason
400	Invalid payload / missing fields
401	Invalid or missing API key
404	Unknown ATM id

7. POST /routing/alternatives

Returns up to 3 alternative operational ATMs. (FR 2, FR 3)

Auth: X-API-Key **NFR:** must respond in under 3s.

Request Body

Field	Type	Required	Description
originAtmId	string	Yes	The failed/out-of-service ATM
latitude	float	Yes	Origin coordinate
longitude	float	Yes	Origin coordinate
requestedService	string	Yes	WITHDRAWAL, DEPOSIT, CURRENCY_EXCHANGE, CHECK_DEPOSIT
requestedAmount	float	Required for WITHDRAWAL	Cash amount needed — filters out ATMs that couldn't fulfill it
radiusKm	float	No (default 5)	Search radius

Cash-Sufficiency Filter (WITHDRAWAL only)

When requestedAmount is present, the Routing Engine excludes any candidate ATM that does not satisfy:

`current_cash_balance >= (requestedAmount + buffer_amount)`

Field	Type	Description
buffer_amount	float (config)	Safety margin kept server-side so the machine isn't drained to exactly zero; not exposed to the client

Response 200 OK — alternatives[]

Field	Type	Description
atmId	string	Alternative ATM id
branchName	string	Display name

Field	Type	Description
distanceKm	float	Distance from origin
etaMinutes	int	Estimated time of arrival
status	string	ONLINE (always, since filtered)
cashStatus	string	AVAILABLE / LOW
availableCashBalance	float	Present only when requestedAmount was sent; confirms the buffer check
distanceSource	string	GOOGLE_MAPS or HAVERSINE_FALLBACK

Response Meta

Field	Type	Description
calculatedAt	ISO8601 string	Server timestamp

Errors

Code	Reason
404	No operational ATM found nearby with enough cash for requestedAmount (or at all, if not sent)
503	Routing engine / mapping service unavailable

8. POST /notifications/sms

Dispatches SMS asynchronously with routing info. (FR 4)

Auth: X-API-Key **NFR:** phone number encrypted in transit, never persisted.

Request Body

Field	Type	Required	Description
phoneNumber	string	Yes	Customer mobile number (E.164 format)
chosenAtmId	string	Yes	Selected alternative ATM
requestedService	string	Yes	Service being confirmed

Response 202 Accepted

Field	Type	Description
taskId	string	Async task identifier
status	string	QUEUED

Errors

Code	Reason
400	Invalid phone number format
404	chosenAtmId not found

9. GET /notifications/sms/{taskId}/status

Polls delivery status of a queued SMS.

Auth: X-API-Key

Path Params

Name	Type	Description
taskId	string	Task id returned from the dispatch call

Response 200 OK

Field	Type	Allowed Values	Description
taskId	string	—	Task identifier
status	string	QUEUED, SENT, DELIVERED, FAILED	Delivery state
sentAt	ISO8601 string / null	—	Timestamp when sent

10. GET /notifications

Lets an admin browse the notification log tied to each ATM — the API-level realization of the ADMIN "monitors" NOTIFICATION relationship in the ERD.

Auth: Bearer (roles: ADMIN, SUPER_ADMIN)

Query Params

Name	Type	Required	Description
atmId	string	No	Filter by ATM
status	string	No	QUEUED, SENT, DELIVERED, FAILED
dateFrom	ISO8601 string	No	Range start
dateTo	ISO8601 string	No	Range end
page	int	No (default 1)	Pagination
limit	int	No (default 20)	Pagination

Response 200 OK — data[]

Field	Type	Description
notificationId	string	Notification id
atmId	string	Related ATM
message	string	SMS body sent (routing link + confirmation)
status	string	QUEUED, SENT, DELIVERED, FAILED
sentAt	ISO8601 string / null	When it was sent

Note: the customer's phone number is intentionally excluded from this response — it is never persisted (NFR 4, Data Minimization).

Response Meta — pagination

Field	Type
page	int
limit	int
total	int
totalPages	int

11. POST /auth/login

Authenticates a bank admin. (NFR 4 — Authentication)

Auth: None

Request Body

Field	Type	Required
username	string	Yes
password	string	Yes

Response 200 OK

Field	Type	Description
accessToken	string (JWT)	Short-lived access token
refreshToken	string (JWT)	Long-lived refresh token
expiresIn	int	Seconds until access token expiry
role	string	ADMIN or SUPER_ADMIN

Errors

Code	Reason
401	Invalid credentials

12. POST /auth/refresh

Auth: Refresh token in body

Request Body

Field	Type	Required
refreshToken	string	Yes

Response 200 OK

Field	Type
accessToken	string (JWT)
expiresIn	int

13. GET /atms

Lists/filters ATMs for the dashboard. (FR 5)

Auth: Bearer (roles: ADMIN, SUPER_ADMIN)

Query Params

Name	Type	Required	Description
status	string	No	ONLINE, OFFLINE, MAINTENANCE
cashStatus	string	No	AVAILABLE, LOW, EMPTY
service	string	No	Filter by supported service
lat	float	No	Geo filter center
lng	float	No	Geo filter center
radiusKm	float	No	Geo filter radius
page	int	No (default 1)	Pagination
limit	int	No (default 20)	Pagination

Response 200 OK — data[]

Field	Type	Description
atmId	string	ATM id
branchName	string	Display name
status	string	ONLINE, OFFLINE, MAINTENANCE
cashStatus	string	AVAILABLE, LOW, EMPTY
services	array<string>	Supported services
latitude	float	Geo coordinate
longitude	float	Geo coordinate

Field	Type	Description
lastHeartbeatAt	ISO8601 string	Last received heartbeat

Response Meta — pagination

Field	Type
page	int
limit	int
total	int
totalPages	int

14. GET /atms/{atmId}

Single ATM detail with recent heartbeat history.

Auth: Bearer (roles: ADMIN, SUPER_ADMIN)

Path Params

Name	Type
atmId	string

Response 200 OK

Field	Type	Description
atmId	string	ATM id
branchName	string	Display name
status	string	Current status
cashStatus	string	Current cash status
services	array<string>	Supported services
latitude	float	Geo coordinate
longitude	float	Geo coordinate
lastHeartbeatAt	ISO8601 string	Last heartbeat time
recentHeartbeats[]	array<object>	Recent status history (status, cashStatus, receivedAt)

Errors

Code	Reason
404	ATM not found

15. POST /atms

Registers a new ATM.

Auth: Bearer, role: SUPER_ADMIN only (RBAC)

Request Body

Field	Type	Required	Description
atmId	string	Yes	Unique identifier
branchName	string	Yes	Display name
latitude	float	Yes	Geo coordinate
longitude	float	Yes	Geo coordinate
services	array<string>	Yes	Initial supported services

Response 201 Created

Returns the created ATM object (same shape as GET /atms/{atmId}).

Errors

Code	Reason
400	Validation error
403	Caller is not SUPER_ADMIN
409	atmId already exists

16. PATCH /atms/{atmId}

Updates ATM metadata (branch name, services).

Auth: Bearer, role: SUPER_ADMIN only

Path Params

Name	Type
atmId	string

Request Body (all optional, partial update)

Field	Type	Description
branchName	string	New display name
services	array<string>	Updated supported services

Response 200 OK

Returns the updated ATM object.

Errors

Code	Reason
403	Caller is not SUPER_ADMIN
404	ATM not found

17. GET /services

Returns the static list of supported service types.

Auth: Bearer or X-API-Key

Response 200 OK — data[]

Field	Type	Description
serviceId	string	Service identifier
name	string	WITHDRAWAL, DEPOSIT, CURRENCY_EXCHANGE, CHECK_DEPOSIT

18. WebSocket /ws/dashboard

Real-time push channel for the Admin Dashboard. (FR 5, NFR 1)

Auth: JWT passed as query param: ?token=<jwt_access_token>

Event: atm.status.updated

Pushed within 2 seconds of a heartbeat state change.

Field	Type	Description
event	string	"atm.status.updated"
payload.atmId	string	ATM id
payload.status	string	ONLINE, OFFLINE, MAINTENANCE
payload.cashStatus	string	AVAILABLE, LOW, EMPTY
payload.latitude	float	Geo coordinate
payload.longitude	float	Geo coordinate
payload.updatedAt	ISO8601 string	Update timestamp

Event: atm.alert

Pushed for state changes needing admin attention.

Field	Type	Description
event	string	"atm.alert"
payload.atmId	string	ATM id
payload.severity	string	LOW, MEDIUM, HIGH
payload.message	string	Human-readable alert text

19. Role-Based Access Control (RBAC) Matrix

Endpoint	ADMIN	SUPER_ADMIN
GET /atms	Yes	Yes
GET /atms/{atmId}	Yes	Yes
POST /atms	No	Yes
PATCH /atms/{atmId}	No	Yes
GET /services	Yes	Yes
GET /notifications	Yes	Yes
WS /ws/dashboard	Yes	Yes

20. Coverage Check — Diagrams to Endpoints

Diagram Element	Covered By
ATM.sendHeartbeat() (Class Diagram)	#2 POST /atms/{atmId}/heartbeat
RoutingEngine.findAlternatives() (Class Diagram)	#3 POST /routing/alternatives
SMSService.sendSMS() / dispatchAsync() (Class Diagram)	#4 POST /notifications/sms
Admin.viewDashboard() / filterATMs() (Class Diagram)	#9 GET /atms
DashboardSocketService.pushStatusUpdate() (Class Diagram)	#14 WS /ws/dashboard
ATM_SERVICE / SERVICE tables (ERD)	#13 GET /services
HEARTBEAT_LOG table (ERD)	#10 GET /atms/{atmId} (recentHeartbeats[])
ADMIN "monitors" NOTIFICATION (ERD)	#6 GET /notifications
"Attempt transaction" -> "Check ATM status" (Sequence Diagram)	#1 POST /atms/{atmId}/transactions/attempt

Diagram Element	Covered By
"Push status update (WebSocket)" (Sequence Diagram)	#14 WS /ws/dashboard
ATM.current_cash_balance field (ERD)	#2 POST /atms/{atmId}/heartbeat (cashBalance), #3 POST /routing/alternatives (cash-sufficiency filter)